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Machine Learning-Based Detection of Wormhole Attacks in IoT Networks Using Classification Models

Manar Almalki, Samah Alajmani

Abstract: ***The widespread adoption of Internet of Things (IoT) networks has introduced new cybersecurity challenges, particularly in the form of wormhole attacks. These attacks pose a significant threat to IoT environments by manipulating network routing without altering packet contents, making them difficult to detect using traditional intrusion detection systems (IDS). This study explores the application of machine learning (ML) techniques for detecting wormhole attacks in IoT networks. The research compares five machine learning classifiers Sparse Representation Classifier (SRC), Multi-Layer Perceptron (MLP), Linear Discriminant Analysis (LDA), Quadratic Discriminant Analysis (QDA), and XGBoost based on metrics such as accuracy, precision, recall, F1-score, and computational efficiency. Data preprocessing techniques were applied to a publicly available IoT dataset to improve the performance of these models. Among the classifiers tested, XGBoost demonstrated superior performance with a detection accuracy of 99.97%, outpacing both traditional and deep learning models. The results highlight the potential of ensemble learning approaches in enhancing IoT security, especially for real-time applications in resource-constrained environments. The study underscores the importance of balancing detection accuracy with computational efficiency when selecting models for dynamic IoT networks. Future work will explore federated learning and hybrid deep learning models to further improve the detection capabilities of wormhole attacks in IoT settings.***

Keywords: ***Anomaly Detection, Cybersecurity, Intrusion Detection, IoT Security, Machine Learning, Wormhole Attacks, XGBoost.***

Abbreviations:
AODV: Ad hoc On-demand Distance Vector
ML: Machine Learning
KNN: K-Nearest Neighbors
GBM: Gradient Boosting Machine
DLBMs: Deep Learning-Based Models
CNM: Conventional Network Monitoring
IQR: Interquartile Range
DT: Decision Trees
DL: Deep Learning
RTT: Round-Trip Time
NB: Naïve Bayes
IDS: Intrusion Detection Systems
SVMs: Support Vector Machines
SGD: Stochastic Gradient Descent
WNCS: Wireless Network Coding Systems
ICMP: Internet Control Message Protocol
MLP: Multi-Layer Perceptron
QDA: Quadratic Discriminant Analysis
SRC: Sparse Representation Classifier
LDA: Linear Discriminant Analysis
MLP: Multilayer Perceptron
FPR: False Positive Rate
MLP: Multi-Layer Perceptron
PR: Precision-Recall

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*****Correspondence Author(s)
Manar Mishal Almalki*, Department of Cybersecurity, Taif University, (Taif), Saudi Arabia. Email ID: S44680292@students.tu.edu.sa, ORCID ID: 0009-0002-0678-5083
Samah Hazzaa Alajmani, Assistance Professor, Department of Information Technology, Taif University, (Taif), Saudi Arabia. Email ID: s.ajmani@tu.edu.sa, ORCID ID: 0009-0000-7152-9559

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I. INTRODUCTION

The growing penetration of the Internet of Things (IoT) in major industries such as health, industrial automation, and smart cities has accompanied remarkable progress in real-time data processing, automation, and decision-making [1]. These advantages are, nonetheless, accompanied by significant security threats, since IoT networks are subject to a number of cyber vulnerabilities, primarily as a result of their distributed nature, limited processing power, and absence of built-in security measures [2]. Wormhole attacks are a serious threat by compromising network integrity through the manipulation of routing paths, thus misleading legitimate nodes to choose malicious routes. This can result in serious disruptions, such as data interception, packet loss, and unauthorized access, which present serious risks to mission-critical applications like medical monitoring systems, smart grids, and industrial control networks [3].

A wormhole attack is when multiple compromised nodes establish a low-latency direct link, or tunnel, between faraway points in the network, tricking routing systems into choosing these compromised routes. Unlike conventional cyberattacks, wormhole attacks do not modify packet payloads; rather, they manipulate network topology, making them extremely hard to detect [4]. Traditional intrusion detection systems (IDS) are based on signature detection and encryption methods; however, these approaches usually prove ineffective against wormhole attacks due to their covert and protocol-independent nature. Therefore, improved detection mechanisms need to be developed to protect IoT networks from such advanced attacks.

In recent years, machine learning (ML) solutions have become plausible alternatives to IoT security, showcasing their capabilities to identify sophisticated patterns of attack, classify network abnormalities, and evolve with emerging cyber threats [5].

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Several works have explored the use of machine learning-based solutions to the detection of wormhole attacks, demonstrating their superiority compared to rule-based and heuristic solutions. Deep learning models and ensemble learning architectures have been particularly helpful in modeling non-linear attack behavior and enhancing detection accuracy. Nevertheless, there remain challenges in identifying the most suitable model that achieves a trade-off among high detection accuracy, low computational cost, and real-time applicability for resource-limited IoT environments [6].

This study seeks to develop and compare machine learning-based detection models for wormhole attacks in IoT networks. We compare five machine learning classifiers, namely Sparse Representation Classifier (SRC), Multi-Layer Perceptron (MLP), Linear Discriminant Analysis (LDA), Quadratic Discriminant Analysis (QDA), and XGBoost, to determine the optimal model for detecting wormhole attacks. The models are evaluated based on accuracy, precision, recall, F1-score, and computational complexity to check their suitability for deployment in low-power IoT networks. The research adds to the literature by offering a detailed comparison of machine learning-based security mechanisms, emphasizing the superiority of ensemble learning and deep learning models in handling sophisticated cyberattacks.

The remainder of this paper is organized as follows: Section 2 is a review of the literature, explaining what has already been done regarding wormhole attack detection and pointing out existing research gaps. Section 3 explains the proposed methodology of detecting wormhole attacks. Section 4 shows the results and discusses the efficiency and effectiveness of the proposed method, while Section 5 summarizes the paper and gives recommendations on the areas of further study.

II. LITERATURE REVIEW

Detection and prevention of wormhole attacks against IoT and ad-hoc networks have been a prominent area of research, resulting in various proposed techniques that include traditional network monitoring techniques along with machine learning (ML) and deep learning (DL) models. Traditional techniques primarily include topology-based, cryptographic, or signal-strength-based solutions. AI-based models now enhance detection effectiveness through automated anomaly detection along with greater scalability within dynamic networks. This section offers a structured overview of the earlier literature, identifies methodological variations, and provides a comparative assessment of the current studies.

A review of different related studies was conducted to build an optimal detection framework for the detection of wormhole attacks on IoT and ad-hoc networks. Comparative research results in Table I relevant to this study are pointed out, highlighting key technologies, contributions, and limitations. It begins with the conventional network monitoring methods, shifts towards machine learning and deep learning-based detection models, and culminates with hybrid approaches utilizing AI-based intrusion detection systems.

A. Traditional Wormhole Attack Detection Approaches

Early studies on the detection of wormhole attacks concentrated on conventional security mechanisms such as round-trip time (RTT) analysis, topology-based approaches, and cryptographic authentication techniques. These methods aimed to detect routing anomalies created by hostile tunnels in the network.

Zaw Tun and Aung Htein Maw (2008) suggested an algorithm for wormhole tunnel detection based on round-trip time (RTT) and neighborhood counts, the goal being the detection of wormhole tunnels based on observation of anomalous delay and abnormal connectivity between nodes in Ad hoc On-Demand Distance Vector (AODV) networks [7]. Results of simulation indicated RTT-based detection of wormhole tunnels is accurate with no special hardware or encryption keys required. The scheme had no ability to detect multi-hop wormhole attacks and had an extremely high dependency on network topology stability, hence reducing its scalability for use in dynamic IoT networks.

Ji (2015) examined wormhole attacks on Wireless Network Coding Systems (WNCS) using measurement of change in the Expected Transmission Count (ETX) metric as a key vulnerability [8]. The paper proposes two novel detection models: a centralized one with trusted control nodes for anomaly detection and a peer-to-peer model (DAWN) on the basis of anomalous packet traffic changes for decentralized detection. 89.43% of detection was done by DAWN; however, the study found some disadvantages relating to collusion resistance in that victim nodes could misrepresent routing parameters to avoid being detected.

Traditional detection mechanisms offer theoretical security enhancements; however, their practical implementation in massive-scale IoT systems is hampered by high false positives, network topology dependence, and computational inefficiencies. Machine learning-based detection models that learn attacks dynamically have been investigated by researchers, thereby providing enhanced scalability and accuracy.

B. Machine Learning-Based Approaches

The application of machine learning models to detect wormhole attacks has increased tremendously, as they can recognize attack pattern in network traffic data and enhance the effectiveness of real-time detection.

Gupta, Singh, and Tiwari (2022) carried out a systematic review of machine learning techniques in wormhole detection and established that Support Vector Machines (SVMs) and Decision Trees (DTs) are the top-performing classification algorithms [9]. The research identified that hybrid approaches that integrate traditional network monitoring techniques with machine learning anomaly detection achieved better detection rates with fewer false positives than standalone cryptographic or topological approaches. The study enumerated significant limitations such as high energy consumption, lengthy processing time, and difficulty in adapting to dynamic network topologies.

Prasad, Tripathi, and Dahal (2019) analyzed machine learning models for detecting wormhole attacks, comparing the performance of Naïve

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Bayes (NB) and Stochastic Gradient Descent (SGD) classifiers [10]. The paper presented a dataset of 20 features, including hop count, packet size, and message type measurements, and showed that SGD performed better than traditional DelPHI-based detection methods, with a detection rate of 93.12%. However, their approach required huge labeled datasets, which was problematic for deployment on real-time IoT networks where data labeling is impossible.

A study on the performance of various machine learning classifiers like K-Nearest Neighbors (KNN), Decision Trees (DT), Convolutional Neural Networks (CNN), and Linear Discriminant Analysis (LDA) [11] was carried out by Abdan and Seno (2022). The outcome revealed that Decision Trees performed optimally with 98.9% accuracy, surpassing CNN with 96.4% and SVM with 98.2%, whereas KNN and LDA were poorer in classification. Despite these enhancements, the study identified real-time adaptability and computational complexity as vital concerns of ML-based wormhole detection systems.

Although ML-based models improve detection rates, their implementation in real-time, resource-constrained IoT networks is made more difficult by their customary requirement for substantial processing resources and big labeled datasets. To solve these obstacles, researchers have turned to deep learning techniques, which provide improved accuracy and flexibility.

C. Deep Learning Approaches for Wormhole Attack Detection

The creation of precise and flexible intrusion detection systems for wormhole attack prevention has been made easier by recent developments in deep learning techniques.

In order to detect wormhole attacks in IoT networks, Abdullah, Albaihani, Osman, and Omar (2024) developed an LSTM-based Intrusion Detection System (IDS) with a 99% accuracy rate, outperforming Decision Trees at 94% and Naïve Bayes at 93% [12]. Their research underscored the significance of temporal pattern recognition in network anomaly detection, noting that LSTM models are particularly effective in identifying attack patterns that change over time. The study noted that LSTM models necessitate greater computational resources, which complicates their real-time deployment in IoT systems. In their AI-based study of wormhole attack detection techniques, Hanif et al. (2022) evaluated deep learning architectures (DLAs) such as hybrid machine learning models (HMLMs), Long Short-Term Memory networks (LSTMs), and Convolutional Neural Networks (CNNs) [13]. Significant issues with DL-based wormhole identification were brought to light by the review, particularly large false positive rates and computational inefficiencies. It suggested the adoption of lightweight neural network architectures and federated learning techniques to improve real-time detection efficiency.

Deep learning-based models (DLBMs) give better detection accuracy; however, there remain huge challenges, including high computationally intensive requirements and large amounts of labeled datasets, that limit widescale deployment. Hybrid detection approaches (HDAs) incorporate machine learning (ML), deep learning (DL), and conventional network monitoring (CNM) techniques as a proposed solution.

D. Hybrid Detection Approaches

To increase detection rate and decrease computational overhead, hybrid detection systems combine more than one security feature. For wormhole and ranking attacks detection in RPL-based IoT networks, Zahra et al. (2022) suggested a hybrid model that employs Gradient Boosting Machine (GBM) for high classification accuracy at low computing costs [14]. The study identified challenges in managing dynamic network topologies, indicating a requirement for enhancements in real-time adaptability, despite its efficiency.

Table-I: Existing Works with Key Technologies and Drawbacks

Authors	Detection Method	Key Contributions	Limitations
Zaw Tun & Aung Htein Maw (2008) [7]	RTT-Based Detection	Identified wormhole tunnels via delay-based anomaly detection	High false positives, ineffective for multi-hop networks
Ji (2015) [8]	ETX-Based Detection	Developed DAWN model (89.43% accuracy) for distorted network coding metrics	Vulnerable to collaborative attacks, limited scalability
Gupta, Singh, and Tiwari (2022) [9]	ML-Based Detection	Identified SVM and Decision Trees as effective classifiers	High energy consumption, dataset dependency
Prasad, Tripathi, and Dahal (2019) [10]	Naïve Bayes, SGD	SGD classifier achieved 93.12% detection rate	Requires large labelled datasets, limits real-time use
Abdan & Seno (2022) [11]	Decision Trees, CNN	DT achieved 98.9% accuracy, outperforming other ML models	High computational complexity
Abdullah, Albaihani, Osman, and Omar (2024) [12]	LSTM-Based IDS	Achieved 99% detection rate, highest precision	High computational cost, limited IoT scalability
Hanif et al. (2022) [13]	CNN, Hybrid ML Models	CNN-based IDS improved classification accuracy	Requires extensive labelled datasets, real-time limitations
Zahra et al. (2022) [14]	GBM-Based Hybrid IDS	Achieved high classification accuracy with low computational cost	Limited adaptability in dynamic networks

III. METHODOLOGY

The model under consideration aims to identify wormhole attacks in IoT networks through network traffic pattern analysis with the help of machine learning. The detection process is systematic in nature, beginning from dataset preparation where inconsistencies and missing values are addressed. Subsequently, data pre-processing is done for improving data quality by deleting outliers and feature normalization. Feature analysis is done in the feature analysis step where the feature correlations of the dataset are analyzed with the goal of optimizing the detection process. Then different machine learning models are built in order to identify network traffic as normal or malicious. Model testing is done using standard performance measures for measuring accuracy and effectiveness. Figure 1 represents the end-to-end process of the proposed detection framework.

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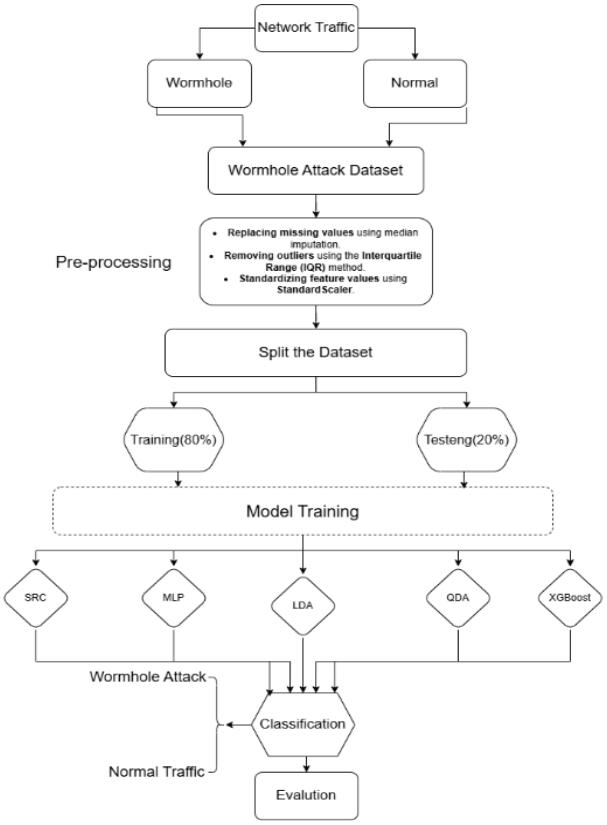

[Fig.1: Proposed Model's Flowchart]

A. Wormhole Attack Dataset

This study employs an openly accessible IoT-based intrusion detection dataset, which has been specifically created to evaluate the performance of machine learning algorithms in detecting different kinds of network intrusions, i.e., wormhole attacks. The dataset contains 637,462 samples out of which 485,334 are labeled as normal traffic and 152,128 as malicious instances in terms of wormhole attacks. Table II provides a concise description of the primary features of the dataset along with their descriptions.

Table-II: Dataset Features and Descriptions

Feature Name	Description	Data Type
duration	Total duration of the packet	float64
Plength	Packet length	int64
Mlength	Maximum length of the packet	int64
HoP	Hop count (number of nodes traversed)	int64
Sno	Sequence number of the packet	int64
Sindex	Source index of the packet	int64
land	Whether the packet has landed	int64
Tmode	Transmission mode	int64
Neighbours	Number of neighbours detected	int64
Hflow	Flow rate of the hops	int64
AvgFlow	Average flow rate across the network	float64
Lflow	Low flow rate	int64
AvgHopCount	Average number of hops per transmission	float64
failedConnection	Number of failed connection attempts	int64
Failed Rate	Rate of failed connections	float64
AODV	Presence of AODV protocol	float64
ICMP	Presence of ICMP protocol	float64
UDP	Presence of UDP protocol	float64
Unknown MsgType	Unknown message type ratio	float64
Route Error	Route error frequency	float64
Route Reply	Route reply frequency	float64
Route Reply Acknowledgment	Route reply acknowledgment presence	float64
Route Request	Route request frequency	float64

B. Data Preprocessing

The dataset underwent pre-processing procedures to maintain data quality and enhance model performance. In order to improve consistency and accuracy in machine learning-based wormhole attack detection, the techniques described address missing values, outliers, and feature scaling.

i. Addressing Missing Values

The dataset's missing values were found and fixed using median imputation. This approach reduces bias brought on by mean imputation or arbitrary value swaps while preserving the dataset's statistical distribution. The dataset is kept balanced and reliable for model training by filling in missing values with the median of each feature.

ii. Detection and Removal of Outliers

Outliers were identified and eliminated through the application of the Interquartile Range (IQR) method. This technique identifies outliers that may adversely affect classification performance by examining the first (Q1) and third quartiles (Q3) of each feature's distribution [15]. Any value falling outside the range defined by:

$$IQR=Q3-Q \quad \ldots \quad (1)$$

$$\text{Lower Bound}=Q1-1.5\times IQR \quad \ldots \quad (2)$$

$$\text{Upper Bound}=Q3+1.5\times IQR \quad \ldots \quad (3)$$

was identified as an outlier and excluded from the dataset. This prevents models from being affected by anomalous data points that do not reflect standard network traffic behavior.

iii. Feature Normalization

The dataset was standardized by scaling numerical features with StandardScaler, which adjusts each feature to possess a mean of zero and a standard deviation of one. This normalization guarantees that all features contribute equally to model training, thereby preventing attributes with larger numerical ranges from over-shadowing the learning process [16]. The transformation is expressed as follows:

$$X_scaled= (X-\mu)/\sigma \quad \ldots \quad (4)$$

where X is the original feature value, μ is the mean, and σ is the standard deviation.

iv. Dataset Splitting Strategy

The features were standardized by scaling numerical features with StandardScaler, which transforms each feature to have a mean of zero and a standard deviation of one. The normalization helps all the features to contribute equally to the model training, thereby preventing attributes with larger numerical ranges from dominating the learning of the model. The transformation is given as:

The dataset was separated into training and testing sets for model generalization assessment. An 80-20 split was used, where 80% of the dataset was allocated for training and 20% for testing. This allows for a sufficient number of samples for training as well as

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evaluation.

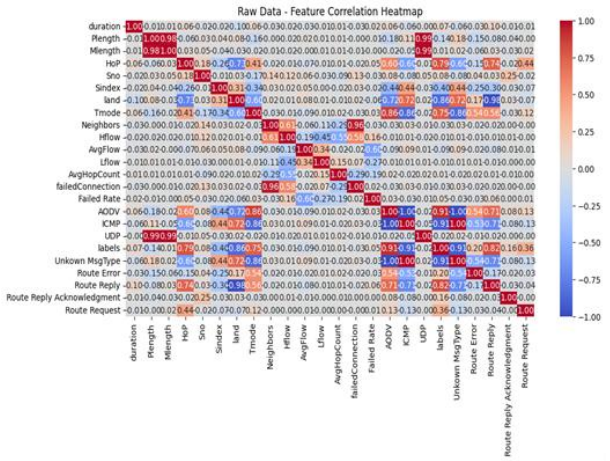
These preprocessing methods ensured that models learn from clean, balanced, and standardized input data, preparing the dataset for efficient machine learning-based lassification.

C. Feature Analysis

Feature analysis plays an important role in determining the importance of various attributes in identifying wormhole attacks in IoT networks. This study departs from standard feature selection by retaining all the features. Instead of removing less significant or redundant attributes, it conducts correlation analysis to comprehend relationships among various attributes. A correlation heatmap was created to observe the relationship between features and verify the correlation among them. Although high-correlation-value features (above 0.8) were found, direct feature removal did not take place, as each feature was determined to be useful for learning complex patterns of network traffic classification.

By preserving all the features of the dataset, the models were given a deep pool of features, and thus they could understand complex interrelations between various network parameters. This helped to avoid loss of vital information and ensured that the models could use all the provided data to increase their predictive performance in differentiating normal traffic from wormhole attack traffic.

Figure 2 illustrates the Correlation Heatmap of Dataset Features. The relationships between the various features in the dataset are shown visually in the correlation heatmap. Stronger relationships are indicated by features with higher correlation values, whilst weaker connections are suggested by features with lower values. This analysis makes sure that all pertinent data points are used efficiently by illuminating the ways in which various features contribute to the identification of wormhole attacks.



[Fig.2: Correlation Heatmap of Dataset Features]

D. Machine Learning Models Used

The models were given a rich feature set by keeping all the properties of the dataset, allowing them to learn complicated relations between different network parameters. By conserving important data, the models were capable of using all available data to improve their prediction accuracy to separate legitimate traffic from the traffic of wormhole attacks.

Based on their performances on anomaly detection and network traffic classification, multiple machine learning models were chosen in detecting wormhole attacks in IoT networks. Compared with pattern detection, feature extractions, as well as computationally efficient analysis, each of these models varies in strengths. Below are detailed descriptions of chosen models.

i. Sparse Representation Classifier

SRC is a classification technique that represents data using a sparse linear combination of basis vectors. This approach finds patterns with less reliance on large datasets, making it particularly useful for anomaly identification. SRC is able to discern between attack and normal traffic with accuracy because to sparse coding techniques [17].

ii. Multi-Layer Perceptron

A feedforward neural network with several layers using non-linear activation functions is called a multilayer perceptron (MLP). This technique is frequently used in intrusion detection and is capable of learning complex decision boundaries [18]. The model is organized in the following manner:

- Input layer: Accepts feature vectors that represent network traffic.
- Hidden layers perform weighted transformations and apply non-linear activations.
- The output layer produces classification probabilities for both normal and attack traffic.

iii. Linear Discriminant Analysis

LDA is a supervised learning technique that improves class separability and data dimensionality reduction. By projecting input features to an optimal subspace, LDA optimizes classification performance, particularly when the dataset feature set has linear relations [19].

iv. Quadratic Discriminant Analysis

By using quadratic decision boundaries to depict class distributions, Quadratic Discriminant Analysis (QDA) improves on Linear Discriminant Analysis (LDA). QDA allows different covariance matrices for each class, which makes it more suitable for datasets with non-linear feature distributions than LDA, which assumes a common covariance structure among classes [20].

v. Extreme Gradient Boosting (XGBoost)

XGBoost is a decision tree-based gradient boosting method designed for high-performance classification applications. By integrating weak classifiers into a robust ensemble model, the learning process is streamlined, improving computing efficiency and accuracy [21]. Notable benefits consist of:

- Utilizing integrated imputation techniques to address missing values.
- Regularization strategies to reduce overfitting.
- Parallelization of execution improves scalability for large datasets.

E. Model Training and Evaluation

To determine how well the selected machine learning models (MLMs) detected wormhole assaults, a systematic training and

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assessment procedure was put in place. The preprocessed dataset was used to train the models, and their classification performance was then evaluated using unseen data.

i. Training Process

In order to obtain adequate data for learning and evaluation purposes, the dataset was split into 80% for training and 20% for testing. Every model was trained using the training set, and its performance was then assessed using the test set. The following steps made up the training process:

- Feature Scaling: Input features were standardized with StandardScaler to ensure numerical consistency.
- Model Fitting: Each model was trained on the labeled training data with optimized hyperparameters.
- The trained models classified test samples into two categories: normal traffic and wormhole attack traffic.
- Performance Evaluation: The outputs of the model were compared with actual labels to calculate evaluation metrics.

ii. Evaluation Metrics

A comprehensive assessment was conducted using multiple evaluation metrics to ensure a balanced analysis of model performance. The metrics chosen are outlined below, accompanied by their corresponding equations [22].

iii. Accuracy

Accuracy measures the proportion of correctly classified instances out of the total dataset. It is given by:

$$\text{Accuracy} = (TP+TN)/(TP+TN+FP+FN) \quad \ldots \quad (5)$$

where:

TP = True Positives (correctly classified attacks).

TN = True Negatives (correctly classified normal traffic).

FP = False Positives (normal traffic misclassified as attacks).

FN = False Negatives (attacks misclassified as normal traffic).

- *Precision*

Precision indicates the proportion of correctly identified attack instances among all instances predicted as attacks. It is defined as:

$$\text{Precision} = TP/(TP+FP) \quad \ldots \quad (6)$$

- *Recall*

Recall (or Sensitivity) measures the model's ability to correctly detect actual attacks. It is given by:

$$\text{Recall} = TP/(TP+FN) \quad \ldots \quad (7)$$

- *F1-Score*

The F1-score is the harmonic mean of precision and recall, providing a balanced evaluation when dealing with imbalanced datasets. It is calculated as:

$$F1 = (2\times\text{Precision}\times\text{Recall})/(\text{Precision}\times\text{Recall}) \quad \ldots \quad (8)$$

- *AUC-ROC Curve*

The Area Under the Receiver Operating Characteristic (AUC-ROC) curve evaluates the trade-off between the true positive rate (recall) and the false positive rate (FPR) across different classification thresholds [23]. A higher AUC score indicates better discrimination between attack and normal traffic instances [24].

- *Computational Time*

To measure the efficiency of each model, computational time was recorded during both training and inference phases [25]. This helps assess the trade-off between model accuracy and processing speed, which is crucial for real-time IoT security applications [26].

IV. RESULTS AND DESSCTION

This section explains a detailed comparison of the models used for detecting worm-hole attacks in IoT networks. The performance of the models was compared using various evaluation metrics, including accuracy, precision, recall, F1-score, AUC-ROC, and computational time. The outcomes were also compared with another research that utilized the same dataset to determine the efficiency of the proposed scheme.

A. Performance Analysis of the Models

To identify the most effective model for wormhole attack detection, five classifiers were evaluated based on their ability to distinguish between normal and malicious traffic. Table III provides an overview of their performance.

Among the models, XGBoost significantly outperformed the others, achieving the highest accuracy (99.97%) while maintaining a strong balance between precision (99.92%) and recall (99.95%). MLP also demonstrated strong performance but required considerably longer computational time, making it less suitable for real-time applications. In contrast, models like LDA and QDA achieved reasonable accuracy but suffered from higher false positive rates, reducing their reliability in real-world scenarios.

Figure 3 presents the confusion matrices for the evaluated models: (a) Confusion matrix for the Sparse Representation Classifier (SRC), illustrating its classification performance and false positive rate; (b) Confusion matrix for the Multi-Layer Perceptron (MLP), highlighting its strong classification accuracy and minimal misclassification errors; (c) Confusion matrix for Linear Discriminant Analysis (LDA), demonstrating its ability to distinguish between normal and malicious traffic, albeit with a higher false positive rate; (d) Confusion matrix for Quadratic Discriminant Analysis (QDA), showcasing its improved recall but lower precision; (e) Confusion matrix for XGBoost, confirming its superior classification accuracy with minimal false positives and false negatives.

To get a clearer picture of our models' performance, we created ROC curves. These graphs show the balance between the true positive rate and the false positive rate. After checking the AUC to pinpoint the best models for classification, we discovered that XGBoost and MLP stood out as the most successful.

The Precision-Recall (PR) curve serves as another crucial tool to evaluate performance. It shows how precision and

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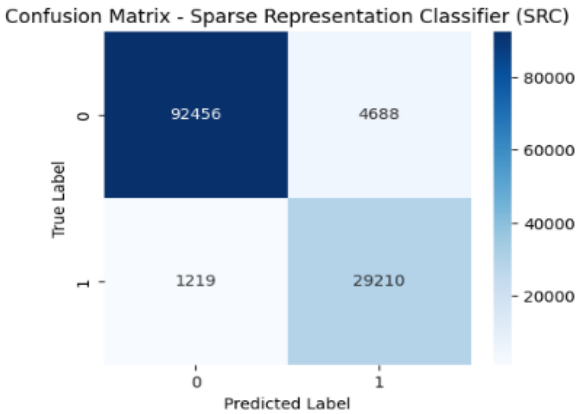
recall balance each other. XGBoost and MLP showed the most steady balance between precision and recall proving they work well to spot wormhole attacks. Check out Figure 4, it really highlights how well our models perform. In part (a), you'll find ROC curves, which give us a clear picture of the balance each model strikes between the true positive rate (TPR) and the false positive rate (FPR). XGBoost and MLP are the real stars here, with their top-notch AUC scores showing they're the most skilled at classification.

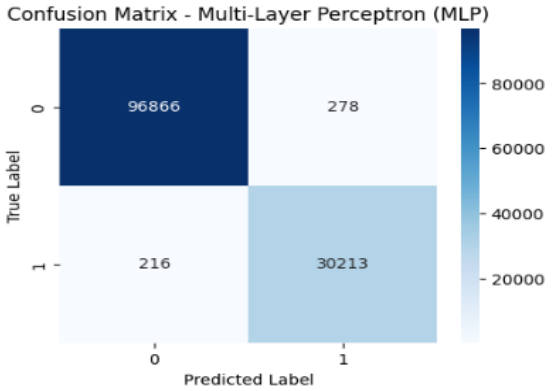
Then, in part (b), we have the Precision-Recall (PR) curves, which help us understand how precision and recall vary among the models. And just like before, XGBoost and MLP come out on top, proving they're exceptional at identifying wormhole attacks.

Table-III: Model Performance Summary

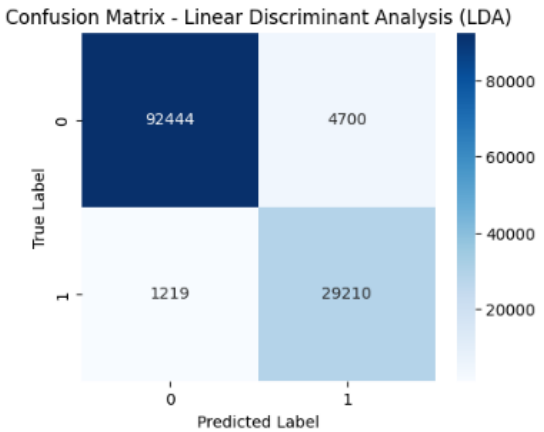
Model	Training Accuracy	Validation Accuracy	Accuracy	Precision	Recall	F1-score	AUC-ROC
SRC	95.36%	95.37%	95.37%	86.17%	95.99%	90.81%	50.00%
MLP	99.51%	99.52%	99.52%	98.91%	99.08%	98.99%	99.98%
LDA	95.35%	95.36%	95.36%	86.14%	95.99%	90.80%	98.99%
QDA	95.57%	95.54%	95.54%	84.50%	99.54%	91.41%	99.43%
XGBoost	99.97%	99.97%	99.97%	99.92%	99.95%	99.93%	99.99%



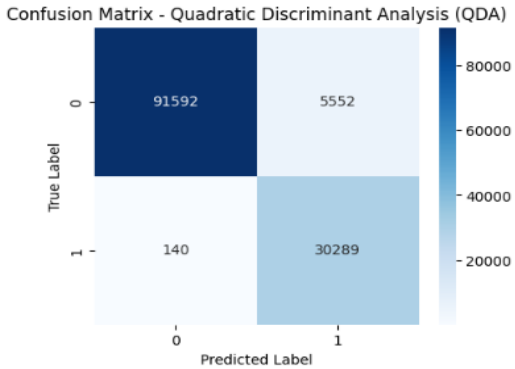
[Fig.3: (a) Confusion Matrix for SRC]



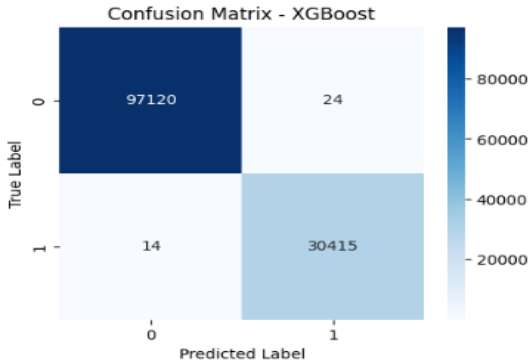
[Fig.3: (b) Confusion Matrix for MLP]



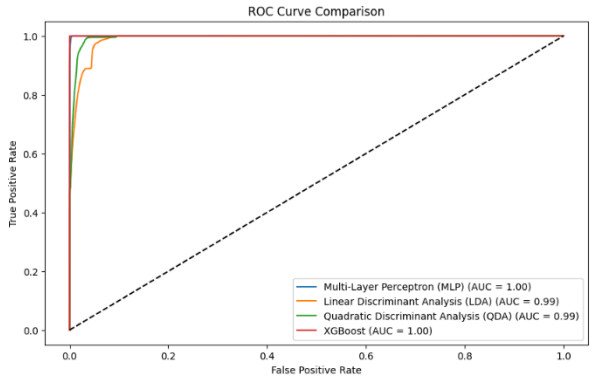
[Fig.3: (c) Confusion Matrix for LDA]



[Fig.3: (d) Confusion Matrix for QDA]



[Fig.3: (e) Confusion Matrix for XGBoost]



[Fig.4: (a) ROC Curve Comparison of All Models]

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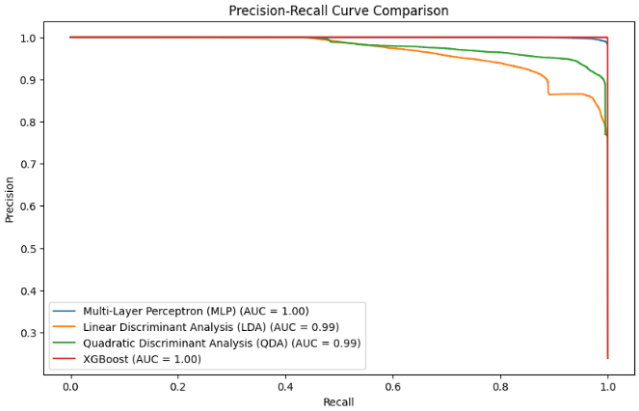

[Fig.4: (b) Precision-Recall Curve Comparison]

Table-IV: Comparison with Other State-of-the-Art Methods

Author	Method-Algorithm	Result				
		Accuracy	Precision	Recall	F1-score	AUC-ROC
Abdullah, Albaihani, Osman, and Omar (2024) [12]	LSTM	99.00%	98.70%	98.40%	98.50%	-
	Decision Tree	94.00%	94.00%	93.80%	93.90%	-
	Naïve Bayes	93.00%	90.50%	91.00%	90.70%	-
Our Proposed Models	SRC	95.37%	86.17%	95.99%	90.81%	50.00%
	MLP	99.52%	98.91%	99.08%	98.99%	99.98%
	LDA	95.36%	86.14%	95.99%	90.80%	98.99%
	QDA	95.54%	84.50%	99.54%	91.41%	99.43%
	XGBoost	99.97%	99.92%	99.95%	99.93%	99.99%

B. Comparison with a Related Study Utilizing the Same Dataset

In order to further support the effectiveness of the proposed strategy, the results obtained were regularly compared with results of a previous study that operated on the same dataset but under different machine learning approaches. Table IV presenting this comparison provides an overall evaluation of the performance of different models on all the significant classification criteria, giving some idea of comparative strengths and limitations of different strategies.

The results indicate that XGBoost outperformed all models from the other study, particularly in terms of accuracy and computational efficiency. While LSTM in the previous study showed strong performance, XGBoost achieved higher accuracy with lower computational requirements, making it a more practical choice for real-time IoT security.

C. Discussion

The findings conclude that XGBoost has the highest performing model in wormhole attack detection between traditional models and deep models in terms of accuracy as well as computational cost. Compared to other methods that used the same dataset, XGBoost had superior classification performance with a very low false positive and false negative rate, thus being highly suitable for practical applications.

Besides, SRC and QDA models were low precision but high recall and thus greater false alarms. It would imply model choice such that precision and recall are proportionate so that there would not be excessive alarms yet attacking accuracy would be retained.

The ability of XGBoost in this study reflects the effectiveness of ensemble learning approaches to securing IoT networks. Being a computationally efficient and highly predictable algorithm, XGBoost provides a stable process for identifying wormhole attacks in adaptive networks.

V. CONCLUSION AND FUTURE WORK

In short, this study successfully developed a machine learning-based detection of wormhole attacks and tested five models to compare their performance in securing IoT networks.

The result indicates that XGBoost outperforms all other models with the best accuracy of 99.97% and therefore the most reliable model to detect wormhole attacks. XGBoost was also seen to achieve improved classification performance with lower computational overhead compared to baseline and deep learning-based models and thus very much appropriate for real-time intrusion detection in resource-constrained IoT networks. The findings of this study are beneficial in enhancing IoT security via the provision of a scalable and efficient detection approach that can effectively identify wormhole attacks with negligible false positives. Comparison with the advanced method used in the reference paper also confirms the efficacy of the proposed method as it sets itself up in relevance with real network security systems. Various potential enhancements can be thought of in the future for increased detection efficiency and dealing with future cyber-attacks.

Integration of deep learning methods such as CNNs or RNNs and XGBoost will once more enhance feature extraction and classification and provide accurate detection.

Coupling real-time anomaly detection with NIDS would allow the model to observe live traffic and thus make it an even more effective means of discovering current attacks. Yet another potential avenue of research is implementing federated learning platforms for enabling models to learn across different IoT devices while ensuring privacy of data and reducing the reliance on centralized processing. Based on these solutions, future research can be optimally tuned and adjusted wormhole attack detection systems to be more adaptive, less prone to false positives, and better immune to sophisticated cyber-attacks in dynamic IoT environments.

DECLARATION STATEMENT

After aggregating input from all authors, I must verify the accuracy of the following information as the article's author.

- **Conflicts of Interest/ Competing Interests:** Based on my understanding, this article has no conflicts of interest.
- **Funding Support:** This article has not been sponsored or funded by any organization or agency. The independence of this research is a crucial factor in affirming its impartiality, as it has been conducted without any external sway.
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approval or participant consent.

- **Data Access Statement and Material Availability:** The adequate resources of this article are publicly accessible.
- **Authors Contributions:** The authorship of this article is contributed equally to all participating individuals.

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AUTHOR'S PROFILE

Manar Mishal Almalki, I hold a bachelor's degree in computer engineering from the College of Computers and Information Technology, Taif University, Taif, SA, and I am currently pursuing a master's degree in cybersecurity. My professional expertise includes understanding security vulnerabilities and exploring methods to enhance system protection. I thrive in diverse and challenging settings, adapting to new technologies and problem-solving in dynamic environments. Committed to continuous learning, I stay updated

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with advancements in cybersecurity to develop my skills and expand my knowledge in the field.

Samah Hazzaa Alajmani, received the B.Sc. degree in 2004 and Ph.D. degree in 2019 in King Abdulaziz University, Jeddah, Saudi Arabia, both in Computer Science. She earned the M.Sc. degree in Information Technology from the Queensland University of Technology, Brisbane, Australia. She is currently an Assistance Professor at Taif University, Taif, Saudi Arabia. Her research interests include Cyber Security, AI, IoT, Deep Learning and Machine learning.

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